

ONAHAMA, Japan

WMO: 475980

Lat: 36.9475N

Lon: 140.9033E

Elev: 5

StdP: 101.26

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.3	-1.3	-11.8	1.4	2.5	-10.6	1.5	2.6	10.0	5.5	9.1	5.4	2.2	340	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	29.2	24.4	28.1	24.1	27.2	23.7	25.4	27.7	24.9	27.2	24.4	26.5	3.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	19.7	26.8	24.2	19.1	26.4	23.7	18.5	26.0	78.2	27.9	76.0	27.3	73.9	26.7	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.3	6.4	DB	-4.3	33.0	1.0	1.6	-5.0	34.2	-5.6	35.1	-6.1	36.0	-6.8	37.1	(4)
			WB	-5.8	26.1	0.8	0.7	-6.3	26.6	-6.8	27.0	-7.3	27.4	-7.8	27.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	4.1	4.3	7.0	11.6	15.8	19.2	22.9	24.8	22.1	17.1	11.5	6.6
	DBStd	7.57	2.13	2.30	2.76	2.91	2.46	2.29	2.19	2.03	2.69	2.75	2.84	2.70
	HDD10.0	592	185	161	102	17	0	0	0	0	0	17	111	
	HDD18.3	2118	443	393	352	203	82	16	1	0	3	57	206	363
	CDD10.0	2042	1	1	8	64	181	277	401	459	363	220	61	6
	CDD18.3	528	0	0	0	0	5	43	143	201	117	19	0	0
	CDH23.3	2474	0	0	0	1	9	67	567	1287	512	31	0	0
	CDH26.7	330	0	0	0	0	1	6	62	204	53	3	0	0
Wind	WSAvg	2.8	3.2	3.2	3.2	3.1	2.8	2.3	2.3	2.4	2.6	2.7	2.7	3.0
Precipitation	PrecAvg	1410	55	57	95	120	143	142	162	130	189	174	89	54
	PrecMax	1936	156	117	166	229	264	286	330	342	485	454	212	139
	PrecMin	934	4	9	22	26	48	37	30	13	14	38	2	1
	PrecStd	197	39	32	39	62	52	60	71	87	103	111	54	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	14.1	17.4	21.1	24.6	27.3	30.1	31.4	29.7	26.1	20.6	17.2
		MCWB	9.1	9.3	12.0	13.7	16.4	19.7	23.9	24.6	23.7	19.7	15.6	13.4
	2%	DB	11.6	12.1	15.2	18.8	22.1	25.1	28.2	29.7	28.1	23.9	19.1	15.0
		MCWB	7.0	8.0	10.8	13.7	16.6	20.5	23.8	24.8	23.7	19.5	14.7	10.6
	5%	DB	10.0	10.6	13.7	17.4	20.7	23.7	27.1	28.6	27.0	22.7	18.0	13.4
		MCWB	5.7	6.5	9.8	13.1	16.4	20.1	23.5	24.7	23.2	18.6	14.4	9.3
	10%	DB	8.6	9.1	12.3	16.3	19.6	22.6	26.1	27.7	26.0	21.6	16.9	11.9
		MCWB	4.5	5.1	8.6	12.5	16.0	19.5	23.1	24.3	22.8	17.8	13.2	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	11.2	13.8	16.6	19.5	22.6	25.4	26.2	25.7	22.3	17.7	15.3
		MCDB	13.0	13.0	15.5	18.6	21.5	24.8	27.8	28.6	27.8	24.1	18.8	16.9
	2%	WB	8.0	8.9	12.0	15.2	18.3	21.6	24.7	25.7	24.9	21.0	16.3	11.7
		MCDB	10.5	11.3	14.3	17.2	20.3	23.8	26.9	28.1	26.8	22.7	17.8	13.7
	5%	WB	6.4	7.1	10.7	14.3	17.5	20.9	24.1	25.2	24.2	19.7	15.2	9.9
		MCDB	9.5	9.9	12.9	16.3	19.4	22.7	26.2	27.6	26.2	21.8	17.2	12.9
	10%	WB	4.8	5.6	9.2	13.4	16.8	20.1	23.5	24.8	23.6	18.5	13.8	8.3
		MCDB	8.1	8.6	11.7	15.6	18.7	21.8	25.5	27.1	25.5	20.9	16.4	11.4
Mean Daily Temperature Range	5% DB	MDBR	8.1	7.9	7.9	7.5	6.4	5.3	5.0	5.0	5.7	7.0	8.3	8.3
		MCDBR	10.2	10.3	9.8	9.0	7.9	6.9	6.3	6.1	6.2	7.5	9.4	10.2
		MCWBR	7.2	7.7	7.4	6.1	4.8	3.7	3.1	2.8	3.4	4.9	6.7	7.5
	5% WB	MCDBR	9.8	9.8	9.2	7.5	6.5	5.8	5.6	5.4	5.1	6.4	8.1	9.4
		MCWBR	7.8	8.1	8.2	6.0	4.7	3.5	3.1	2.8	3.2	5.1	7.0	7.9
Clear-Sky Solar Irradiance	taub	0.378	0.429	0.506	0.561	0.579	0.605	0.606	0.589	0.516	0.473	0.419	0.382	
	taud	2.209	2.060	1.889	1.815	1.825	1.831	1.883	1.920	2.077	2.106	2.171	2.223	
	Ebn at Noon	784	782	755	739	734	713	709	709	741	730	728	749	
	Edh at Noon	108	141	184	209	210	209	197	186	151	134	112	100	
All-Sky Solar Radiation	RadAvg	2.51	3.15	4.02	4.75	5.17	4.82	4.80	4.77	3.83	3.02	2.40	2.16	
	RadStd	0.14	0.26	0.23	0.43	0.51	0.36	0.75	0.56	0.32	0.29	0.18	0.14	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	+0.52	N/A	N/A	N/A	N/A
	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page